

CAIRO

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HISTORY, CULTURE, AND TRADITIONS

XR Ancient Escape Room Experience



ORIGIN OF THE CREATIVE IDEA

The idea behind the project emerged from a desire to create an immersive and engaging experience inspired by ancient environments such as temples and hidden chambers. It was influenced by the fascination with mystery, exploration, and the challenge of uncovering secrets, where users are encouraged to interact with their surroundings and solve puzzles to progress. The concept also draws from the appeal of escape rooms, combining problem-solving with storytelling in a virtual setting. Additionally, the project was shaped by the challenge of designing intuitive XR interactions, motivating the development of creative solutions that enhance user engagement, immersion, and overall experience



FOCUS ON

1

Exploration

2

Interaction

3

Immersion

- GOAL : CREATE A SENSE OF DISCOVERY
AND CHALLENGE



USER JOURNEY

- 1) Start in Lobby (Arcade themed)
- 2) Interact with Main Menu
- 3) Teleport into Puzzle Chamber
- 4) solve puzzles:
 - Puzzle 1 → Door unlocks
 - Puzzle 2 → Door unlocks
 - Puzzle 3 → Escape

ENVIRONMENT DESIGN (SPATIAL UX)

- Ancient visuals:
 - Stone walls
 - Torches / warm lighting
 - Hieroglyph-style elements
- Design principles:
 - Realistic scale
 - Depth & spatial awareness
 - Guided exploration using lighting

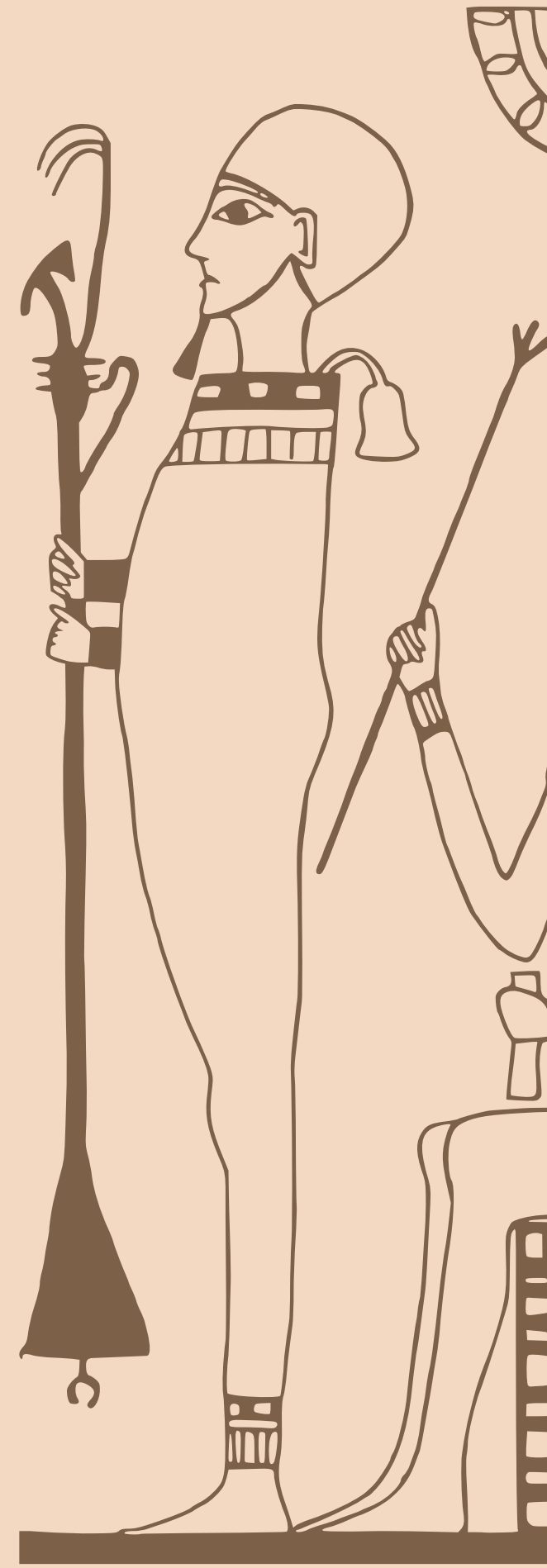


INTERACTION MODEL

- Teleportation (locomotion)
- Object grabbing (direct manipulation)
- Hover interaction
- Button clicking

Tools:

- XR Interaction Toolkit
- Raycasting & Colliders





PUZZLE 1 HIDDEN ARTIFACTS

- Find 3 hidden cubes (artifacts)
- Place them on ancient shelves
- Mechanics:
 - Exploration + grabbing
- Feedback:
 - Audio + visual confirmation
 - Door opens



PUZZLE 2 : ANCIENT BALANCE

- Place:
 - Gold sphere
 - Silver sphere
- Concept:
 - Balance scale puzzle
- Features:
 - Physics-based interaction
 - Real-time feedback



PUZZLE 3 : HIDDEN CODE

- Code appears only when hovering
- Player must:
 - Discover → Memorize → Input
- Interaction:
 - Button sequence input



Feedback & Immersion

- Audio:
 - Ambient ancient sounds
 - Puzzle success sounds
- Visual:
 - UI updates
 - Door animations

Dynamic UI

- Updates with progress:
 - “Puzzle Locked”
 - “Puzzle Completed”
- Positioned in 3D space for visibility



Human Factors & Comfort

- Teleportation → reduces motion sickness
- Stable camera
- Clear interaction zones
- No sudden movements

Accessibility

- Simple interactions
- Clear instructions
- Easy navigation

TECHNICAL IMPLEMENTATION

1. Unity + XR Toolkit

2.

- Interaction system
- Puzzle logic

3.

- UI system
- Feedback System

